EyeSense: Mobile Eyesight Measurement via Time-of-Flight Phase Manipulation

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***Abstract*—Over 2.2 billion people worldwide live with vision impairment, and an estimated 80% of cases are preventable or treatable with early diagnosis, yet objective refraction remains largely inaccessible in rural and resource-poor settings due to the cost ($5,000–$20,000+) and operator training required by clinical autorefractors. We present EyeSense, a system for smartphone-based eyesight screening that exploits the infrared (IR) Time-of-Flight (ToF) sensor already embedded in many commodity smartphones to perform eccentric photorefraction without any external hardware. Building on the open-source Mozart ToF framework, we modify the sensor pipeline and build an app, EyeSense, that captures IR phase maps of the retinal reflex. We collect two datasets, one being an 80-participant lab dataset with self-reported prescriptions and a 40-participant clinical dataset from a LUMS eye camp with autorefractor-verified ground truth. We evaluated a Ridge regression baseline, a CNN (ResNet-18) baseline, an augmentation ablation, and a combined Ridge and CNN ensemble for predicting spherical refractive error directly from Mozart phase maps. Our ensemble model achieves an MAE of 1.15D and 64.4%/76.2% accuracy within $\pm$1.0D/$\pm$1.5D of ground truth. We also characterize a distance-dependent sign inversion of the retinal reflex below 20–25cm, caused by the transition from diffuse retinal reflex to specular corneal (Purkinje) reflection.**

***Index Terms*—Time-of-Flight sensing, eccentric photorefraction, mobile health, refractive error screening, infrared imaging, deep learning**

I. Introduction

Uncorrected refractive error is one of the most common, and most treatable, causes of vision impairment globally. More than 2.2 billion people live with some form of vision impairment, and at least 1 billion cases could have been prevented or remain unaddressed [1]. In practice, however, objective refraction, which is the clinical measurement of a patient's prescription without relying on subjective patient feedback, is concentrated in settings with trained optometric staff and dedicated equipment. Autorefractors, the clinical gold standard, cost $5,000–$20,000 per unit and require a trained operator, placing them out of reach for rural clinics, school screening programs, and resource-constrained health systems. Access to objective refraction in these settings is limited, motivating portable screening approaches, including smartphone-based photoscreening [2].

This gap has motivated work on smartphone-based photorefraction, which seeks to capture the clinical retinoscopy/photorefraction signal using mobile imaging hardware [3]–[5]. Eccentric photorefraction, the underlying optical principle, illuminates the eye from a point slightly offset from the camera's optical axis; the eye's refractive state determines the width, shape, and orientation of a bright crescent that forms in the pupil due to the retinal reflex. A perfectly emmetropic eye produces a uniformly illuminated pupil, while myopic and hyperopic eyes produce characteristic crescents whose width is approximately proportional to the magnitude of the refractive error in diopters.

The central obstacle to doing this with a standard smartphone camera is light itself. Visible-light flash photography triggers the pupillary light reflex, constricting the pupil and altering the very signal being measured, and also requires cooperative, often uncomfortable, direct viewing of a bright flash by the subject (an acute problem for the pediatric populations that most need vision screening). Infrared (IR) illumination avoids the pupillary reflex, since the eye is not photopically sensitive to it, but conventional IR photorefraction systems described in the literature achieve this with custom hardware: LED rings, handheld attachments with dedicated IR optics, or bespoke multi-LED illumination arrays. All of this hardware would again be unattainable in rural settings.

Our key observation is that an increasing number of commodity smartphones already contain an IR Time-of-Flight (ToF) depth sensor for purposes such as portrait-mode depth estimation and face unlock. A ToF sensor illuminates the scene with IR light and measures the phase shift of the reflected signal to recover per-pixel depth. Critically, this gives such phones *built-in, eccentric IR illumination* co-located with a rear-facing sensor which is the same geometry that eccentric photorefraction requires. This paper presents EyeSense, a system that repurposes a smartphone's ToF sensor for eccentric photorefraction, and a supervised learning pipeline that maps the resulting phase maps to refractive error (sphere).

Our contributions are:

- We identify and motivate the ToF sensor, rather than RGB or dedicated IR hardware, as a uniquely suited, zero-marginal-cost substrate for mobile eccentric photorefraction, and characterize the optical regime (including a distance-dependent corneal/retinal reflex sign inversion) under which it is usable.
- We build on the open-source Mozart ToF framework and correct a normalization artifact (a fixed 100mm depth floor that clamped and blacked out near-field pixels) with a per-frame adaptive normalization scheme, and implement the EyeSense app with selectable Mozart/Depth/Confidence visualizations.
- We collect and release two datasets: an 80-participant lab dataset with self-reported prescriptions, and a 40-

participant dataset from a LUMS eye camp with professional autorefractor ground truth spanning ages 7–60.
- We evaluate a progression of models: Ridge regression on handcrafted features, a CNN baseline, an augmentation ablation, and a combined Ridge and CNN ensemble. We report accuracy against the diopter error thresholds used in the eyesight screening literature.

The remainder of this paper is organized as follows. Section II reviews the clinical basis for objective refraction and prior smartphone- and custom-hardware-based photorefraction systems, and motivates ToF as an under-explored sensing modality. Section III describes our modifications to the Mozart ToF pipeline, the resulting EyeSense app, an optical limitation we observed at close range, and our two data collection phases. Section IV presents our machine learning pipeline and results. Section V discusses limitations, Section VI outlines future work, and Section VII concludes.

II. Related Work

A. Clinical Optical Diagnostic Architectures

Three optical techniques underlie essentially all objective refraction, clinical or mobile. *Autorefractors* project IR light onto the retina and use the returned wavefront to compute an objective, automated measurement of refractive error; they are the clinical gold standard but require a trained operator and cost $5,000–$20,000+ per unit. *Retinoscopy* is a lower-cost clinical alternative in which a clinician shines a streak of light into the eye and observes the direction of movement of the pupillary reflex, using trial lenses to neutralize (and thereby determine) the refractive error; it remains operator-dependent and slow [6]. *Eccentric photorefraction* is the photographic analog of retinoscopy: a camera captures the retinal reflex directly, and the refracting eye produces a bright crescent in the pupil whose width is proportional to the magnitude of the refractive error [7], [8], as illustrated in Fig. 1. Eccentric photorefraction is the principle EyeSense builds on, because unlike the first two techniques it is passive, single-shot, and camera-native.

B. Smartphone and Custom-Hardware Photorefraction

Chun et al. [3] trained a ResNet-18 classifier on 305 visible-light eccentric photorefraction images of Korean children, achieving 81.6% accuracy across seven refractive-error bins, and used Grad-CAM to identify the image regions driving classification. Their approach is constrained by its reliance on a visible-light flash (inducing pupillary constriction), a small and single-ethnicity dataset, and the absence of any objective phase-level signal.

Linde et al. [4] used a handheld IR fundus camera with dedicated IR optics (a proprietary device referred to as ”nun IR” in project materials) to capture red-reflex images from 512 patients across India and New Zealand, and applied EfficientNet/Inception-v3 to reach 75% accuracy for spherical power in a binary myopic/non-myopic setting. Their approach requires dedicated, non-commodity hardware, provides no raw phase data (only processed IR images), and its cylinder prediction and specificity (47%) are poor.

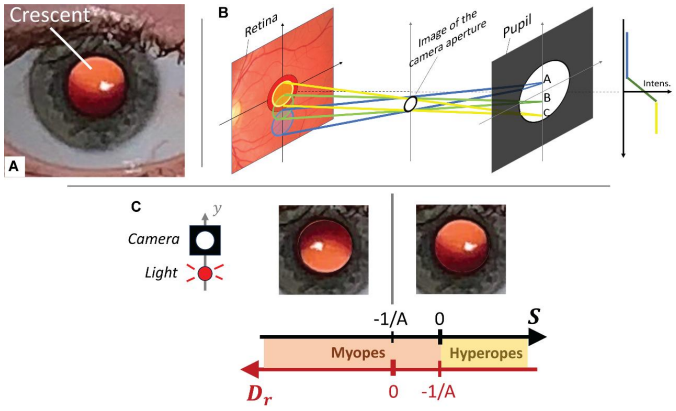


Fig. 1. Eccentric photorefraction optics: (A) the pupillary crescent; (B) the retina-pupil-camera ray geometry; (C) crescent position for myopic vs. hyperopic eyes. Reproduced from Bharadwaj et al. [8].

Xu et al. [5] proposed REDNet, which combines CNN feature extraction with recurrent fusion across multiple eccentric photorefraction images to estimate spherical and cylindrical power. Its multi-view design demonstrates the value of combining information across illumination meridians, while relying on a dedicated photorefractor rather than a commodity phone.

Linde et al. [4] evaluated color and infrared pupil red-reflex images captured using a smartphone-based fundus camera. Their results support IR imaging for refractive-error estimation, while also illustrating that the image pipeline and capture hardware differ from phase-resolved ToF sensing.

Prior work therefore spans visible-light smartphone photorefraction and IR red-reflex systems that use dedicated capture hardware [3]–[5]. To our knowledge, none uses a commodity smartphone ToF sensor to obtain an IR phase-resolved signal for refractive-error estimation.

C. Sensing Modality: RGB, IR, and Time-of-Flight

The modality used to illuminate and image the eye determines which of the above constraints a system inherits. RGB cameras rely on visible light, which triggers the pupillary light reflex, constricting the pupil and confounding the retinal reflex measurement with an active neurological response rather than a static optical one. IR illumination avoids this reflex, since the eye is not photopically sensitive to it [9], and is the basis of every prior smartphone-adjacent photorefraction system reviewed above; however, all of these systems still require external hardware, LED rings, handheld IR attachments, or custom multi-LED arrays, to generate that IR illumination on a conventional smartphone camera. This added hardware cost and setup complexity is the primary barrier to deployment at scale for IR-based approaches.

ToF sensors close this gap directly: they measure per-pixel depth by illuminating the scene with IR light and measuring the phase shift of the returned signal, which means

a ToF-equipped smartphone already contains an eccentric IR illuminant (the ToF emitter, offset from the ToF/RGB sensor) co-located with a sensor capable of resolving fine-grained IR return signal, exactly the geometry eccentric photorefraction requires, with zero additional hardware. Phase-manipulated ToF sensing has separately been shown to recover fine surface texture from commodity smartphone ToF sensors in low-light conditions [10]; EyeSense applies this same phase-manipulation principle to the eye, using it to recover the retinal reflex rather than general scene texture.

III. System and Data Collection

Fig. 2 gives an overview of the full EyeSense pipeline, from raw ToF sensing through to a predicted sphere value, which the remainder of this section and Section IV describe in detail.

A. Mozart Sensor Pipeline Modifications

EyeSense is built on top of Mozart, an open-source framework for accessing raw ToF sensor data on supported Android devices. We identified and corrected a normalization artifact in Mozart's `normalizeRange()` function: it used a fixed `RANGE_MIN = 100mm` floor, so that any pixel with a measured depth below 100mm was clamped and rendered black. Because the eye and face are frequently well within 100mm of the sensor during a photorefraction capture, this clamping discarded exactly the near-field content EyeSense depends on. We replaced this with per-frame adaptive normalization,

$$\text{cutDepth} = \frac{\text{depthF[pixel]}}{\text{maxDepth}}, \tag{1}$$

where maxDepth is the maximum depth value observed within that specific frame, so that every captured frame is normalized against its own real content rather than a fixed global floor.

The EyeSense app (built on this corrected pipeline) provides three selectable visualizations at capture time: *Mozart* (the N2 scalar map used for our ML pipeline), *Depth* (a jet colormap of raw depth), and *Confidence* (a grayscale confidence map), alongside a live numeric depth readout in millimeters, as shown in Fig. 3.

B. The Distance Limitation

Even after normalization was corrected, we observed a consistent behavior as the phone is brought closer than approximately 20–25cm to the eye: the pupil region's brightness inverts, with what should be a bright retinal reflex instead appearing dark relative to the surrounding iris and sclera. We attribute this to a shift in the dominant optical signal at the pupil: below this distance threshold, the specular corneal reflection (the P1 Purkinje image) dominates over the diffuse retinal reflex that eccentric photorefraction depends on. This sets a practical minimum working distance for EyeSense capture and is a constraint that we acknowledge.

TABLE I
Dataset Summary Across Both Collection Phases

	Phase 1 (Lab)	**Phase 2 (Eye Camp)**
Participants	80	40
Frames captured	160	80
Ground truth	Self-reported Rx	Autorefractor
Age range	Not recorded	7–60
Refractive range (sphere)	-9D to $+2$D	-3D to $+3$D
Additional data	–	Cylinder, axis

C. Data Collection: Phase 1 (Lab Dataset)

We conducted an initial data collection round in a controlled-lighting IoT lab setting. Each of 80 participants was photographed with the Mozart-modified ToF app on a single test device, yielding 160 total frames (one per eye). Each participant self-reported their sphere prescription. This yields a dataset spanning a refractive range of -9D to $+2$D sphere, with the caveat that ground truth is self-reported and unverified by a clinician.

D. Data Collection: Phase 2 (Clinically Verified Dataset)

To address the ground-truth limitation of Phase 1, we ran a second collection round at a LUMS eye camp staffed by a professional optometrist with an autorefractor. This yielded 40 participants (80 frames) spanning ages 7–60 and a refractive range of -3D to $+3$D sphere. This phase eliminates self-report noise and additionally provides cylinder and axis ground truth, which Phase 1 lacked.

Table I summarizes both collection phases, and Fig. 4 shows representative crops spanning the refractive range.

IV. Machine Learning Pipeline

A. Problem Formulation and Evaluation Metrics

We treat sphere prediction as a regression problem: given a Mozart phase map for one eye, predict the sphere component of that eye's refractive error in diopters. Following the conventions of prior photorefraction work (Section II-B), we report both regression error (mean absolute error, MAE, and root-mean-square error, RMSE, in diopters) and threshold accuracy (the fraction of predictions within $\pm$1.0D and $\pm$1.5D of ground truth). We adopt $\pm$1.0D as an approximate acceptable screening threshold and $\pm$1.5D as a looser bound, noting that clinical prescription precision is $\pm$0.25D.

B. Phase 1: Ridge Regression Baseline

We first established whether Mozart phase maps carry any signal correlated with refractive error at all, using a deliberately simple, interpretable model. We extracted 8 handcrafted features per image (crescent asymmetry, center-edge intensity ratio, local gradient statistics, and global intensity statistics) and fit a Ridge regression model [11], evaluated with leave-one-out cross-validation (LOO). This confirmed that Mozart maps linearly correlate with prescription, with the model correctly identifying mild myopia at 83.3% accuracy, establishing

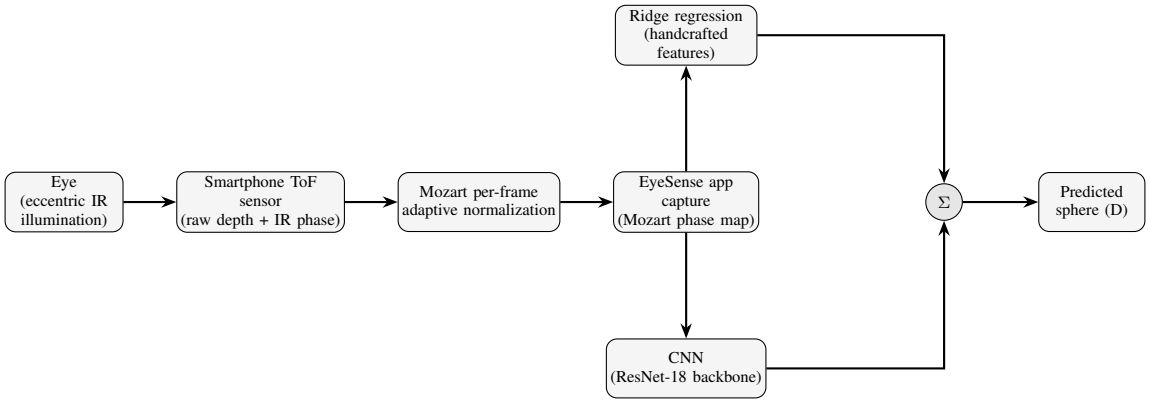


Fig. 2. Overview of the EyeSense pipeline, from raw ToF sensing to a predicted sphere value.



(a) Original Mozart (b) EyeSense

Fig. 3. Original Mozart output vs. EyeSense on the corrected pipeline: adaptive normalization recovers near-field detail, including the retinal reflex, that Mozart's fixed-floor normalization clamped to black.



(a) Emmetropic (b) +1.5D (c) -5D (d) -9D

Fig. 4. Representative labeled Mozart phase-map crops, spanning the refractive range from emmetropic to high myopia.

that the underlying signal is real before committing to more complex modeling.

C. Phase 2: CNN Baseline

We then trained a ResNet-18-backbone [12] CNN end-to-end on raw Mozart phase maps, using 5-fold cross-validation without data augmentation, Adam optimization [13] at a learning rate of 1×10^{-4}, and early stopping with a patience of 20 epochs.

D. Phase 3: Augmentation Ablation

We tested a standard augmentation suite (horizontal flip, ± 0.3 color jitter, $\pm 10^{\circ}$ rotation, Gaussian blur) on top of the Phase 2 CNN. This *degraded* performance, to an MAE of 1.26D versus the non-augmented baseline. We attribute this specifically to horizontal flip: the crescent asymmetry in a Mozart map is a physically directional signal tied to the eccentric illumination geometry, and flipping the image inverts this asymmetry without inverting the associated label, injecting label noise rather than useful invariance.

TABLE II
MODEL PERFORMANCE SUMMARY (SPHERE PREDICTION)

Model	MAE	RMSE	$\pm$1.0D	$\pm$1.5D
Ridge (LOO)	1.286	1.898	53.8%	70.6%
CNN Baseline	1.186	1.950	63.1%	75.6%
CNN w/ Aug.	1.259	2.041	60.6%	70.0%
Ensemble	1.151	1.865	**64.4%**	**76.2%**

E. Phase 4: Combined Ridge and CNN Ensemble

Because the Ridge and CNN models make errors on different subsets of images (their failure modes are only partially correlated), we combined them in a weighted ensemble with a 0.4/0.6 (Ridge/CNN) weighting, chosen so that the CNN carries the majority of predictive weight while Ridge stabilizes predictions in the tails of the refractive-error distribution.

F. Results

Table II summarizes performance across all four modeling stages. The ensemble achieves the lowest MAE (1.151D) and the highest accuracy within both $\pm$1.0D (64.4%) and $\pm$1.5D (76.2%). The CNN baseline has the lowest RMSE (1.950D versus 1.865D for the ensemble).

Fig. 5 plots predicted vs. ground-truth sphere for the Ridge, CNN baseline, and their ensemble. Qualitatively, the Ridge model's predictions cluster toward mild myopia and underpredict the magnitude of high-myopia cases (ground truth beyond -5D), consistent with a linear model's limited capacity to capture the nonlinear relationship between crescent geometry and severe refractive error. The CNN model shows the opposite tendency at the extremes, occasionally over-predicting myopia magnitude on emmetropic (near-zero) ground-truth eyes. The ensemble partially, but not fully, cancels both failure modes.

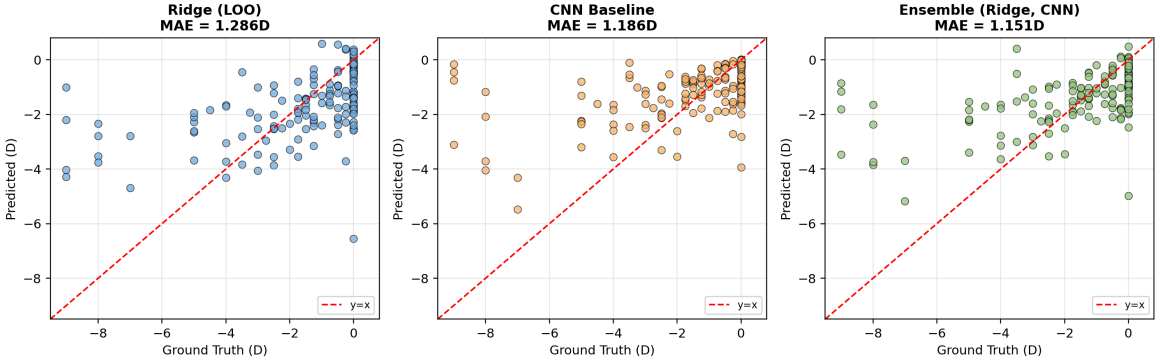


Fig. 5. Predicted vs. ground-truth sphere for each model (LOO cross-validation), with the $y = x$ line marking perfect prediction.

V. Discussion and Limitations

The central limitation of the current pipeline is that both the CNN and the ensemble remain black-box gradient-based estimators of a signal, crescent width and shape, that has a known, direct geometric relationship to refractive error in the eccentric photorefraction literature. This means prediction error is entangled with model capacity and training-data coverage rather than being attributable to a specific, correctable measurement step. Dataset size (240 frames total across both phases) is also small relative to the label-noise floor introduced by the Phase 1 self-reported ground truth, and the two datasets differ in refractive range, age distribution, and label quality in ways that a pooled 5-fold split does not fully account for. The 20–25cm minimum working distance imposed by the corneal/retinal reflex inversion (Section III-B) is also a usability constraint that has not yet been evaluated with real users self-positioning a device, as opposed to a lab operator holding it at a controlled distance.

VI. Future Work

A. Direct Crescent Width Measurement (Geometric N1)

We plan to move from gradient-based CNN inference to direct geometric measurement of crescent width from the ToF phase data, since all of the relevant geometric quantities (crescent width, pupil boundary, illumination offset) are, in principle, directly measurable from the phase map itself. This would eliminate the CNN as a black box for the core refraction calculation and make errors directly attributable to specific measurement steps.

B. Astigmatism Axis from Crescent Orientation (N2)

A second direction is precise angular measurement of crescent tilt to directly recover cylinder and axis, which would allow EyeSense to output a full sphere+cylinder+axis prescription without requiring any patient-reported input, closing the gap with clinical autorefractors.

C. Dataset Expansion and Generalization

We plan to expand data collection to 1,000+ participants spanning a wider range of ages and iris colors, with particular emphasis on a pediatric cohort, since childhood vision screening is the primary intended use case for a zero-cost, hardware-free refraction tool.

VII. Conclusion

We presented EyeSense, a system that repurposes the IR Time-of-Flight sensor already present in commodity smartphones to perform eccentric photorefraction without any external hardware, closing a gap in the prior literature between smartphone-only approaches (limited to visible light and coarse accuracy) and hardware-accurate approaches (which require bespoke, non-commodity illumination arrays). Building on a corrected version of the open-source Mozart ToF pipeline, we collected two datasets, spanning 120 participants and both self-reported and clinically verified ground truth, and showed that a combined Ridge and CNN ensemble predicts spherical refractive error from ToF phase maps with an MAE of 1.15D and 76.2% accuracy within $\pm$1.5D. We further identified a practical minimum working distance imposed by a corneal/retinal reflex transition, and outlined a path toward direct geometric measurement of crescent width and orientation to eliminate the CNN as a black box and recover full sphere-cylinder-axis prescriptions.

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